KIDNEY X

EMPOWER CHALLENGE

Track B: Scalable Prototypes

Solutions ready for pilot testing.

LIVINGLINK

Open-source support for the living kidney donor journey

TRACK B: SCALABLE PROTOTYPE

LivingLink

Open-source support across the entire living kidney donor journey

FUNCTIONAL BETA • PILOT READY

THE PROBLEM & SOLUTION

THE CORE PROBLEM

Willing donors face disconnected information, financial uncertainty, isolation, evaluation delays, and weak long-term follow-up. Each handoff creates another opportunity to withdraw.

OUR NOVEL APPROACH

One donor-centered platform combines five modules: ReadyCheck, DonorShield, MentorMatch, CenterFlow, and LifeAfter. Five public tools remove the sign-up wall; role-based workflows connect donor and center tasks.

OUR TEAM & CAPABILITY

- Project lead: Richard Horsley, submission lead and prototype owner.
- Repository capability: Next.js/TypeScript, Prisma/PostgreSQL, donor workflows, FHIR integrations, AI-assisted experiences, and cloud deployment.
- Pilot partners: two transplant centers to be recruited; no institutional pilot partner is represented as confirmed.
- Pilot support model: donor and coordinator onboarding, test-data setup, workflow training, and monthly outcome review.

MEASURABLE VALUE • 12-MONTH PILOT TARGETS (NOT YET CLINICALLY VALIDATED)

26% → 75%	−20%	−6 weeks	59% → 85%
NLDAC COMPLETION	DONOR WITHDRAWAL	EVALUATION DURATION	TWO-YEAR FOLLOW-UP

WHY IT CAN SCALE

- MIT-licensed code and a single web application reduce deployment complexity and vendor lock-in.
- Public education works without an account; authenticated workflows add role, center, consent, and patient-context boundaries.
- No specialized hardware. Proposed pilots validate workflow fit, user adoption, interoperability, and measurable outcomes.

Evidence of Efficacy & Prototype Showcase

EVIDENCE OF POTENTIAL EFFICACY

EVIDENCE-BASED DESIGN

- Peer education is backed by a submission-cited randomized trial reporting 23% higher donation intent among ambivalent candidates.
- DonorShield puts wage estimates and NLDAC guidance before clinical steps, addressing financial uncertainty early.
- CenterFlow flags evaluations after 14 days without progress; LifeAfter pairs scheduled check-ins with PHQ-2 escalation at scores ≥3.
- Four pilot outcomes are pre-specified: NLDAC completion, withdrawal, evaluation duration, and two-year follow-up.

FUNCTIONAL BETA SURFACES

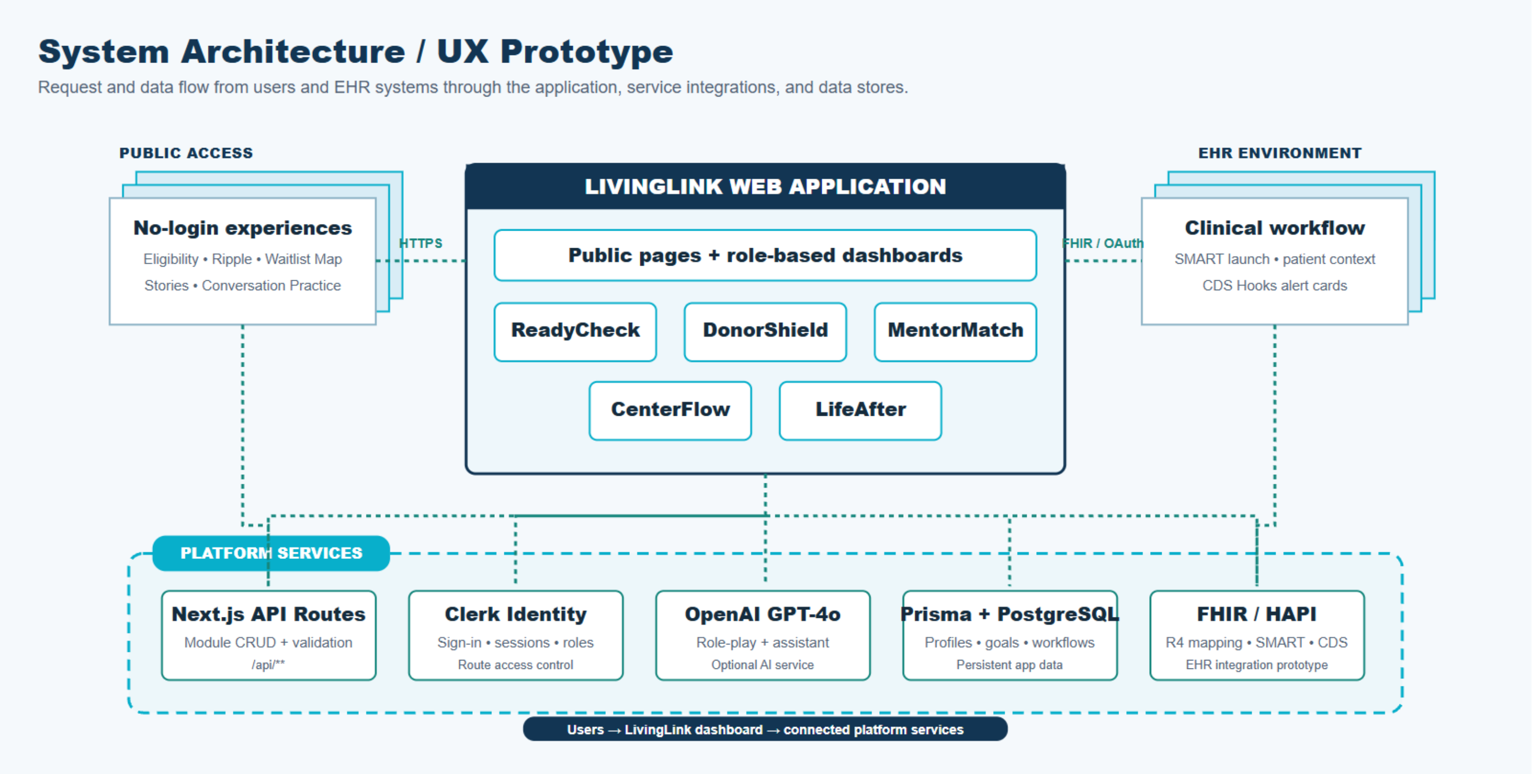
- 5 public experiences: eligibility pre-screen, ripple calculator, waitlist map, stories, AI conversation practice.
- 5 authenticated modules plus donor, coordinator, clinician, patient, and administrator views.
- Prisma-backed APIs for goals, expenses, NLDAC, mentoring, evaluations, check-ins, consent, privacy, and alerts.

No pilot outcome or clinical efficacy is claimed yet.

TECHNICAL INNOVATION & AI USE

FHIR R4 mappings, SMART launch with PKCE, CDS Hooks cards, and structured export are implemented as prototypes. Optional GPT-4o powers public conversation role-play and contextual assistance, while the core donor workflows remain available without AI.

SYSTEM ARCHITECTURE / UX PROTOTYPE



PILOT READINESS

Path to Pilot Implementation

TECHNICAL FEASIBILITY

- Single Next.js 15 / TypeScript application with server route handlers.
- Prisma + PostgreSQL data layer; Clerk integration and role/center authorization helpers.
- Local HAPI FHIR R4 plus FHIR mapping, export, SMART, and CDS Hooks prototypes.
- Unit, integration, end-to-end, and accessibility test assets support iterative pilot preparation.

INTEROPERABILITY

- FHIR R4 resources: Patient, Observation, Goal, Coverage, Communication, Task, CarePlan, and QuestionnaireResponse.
- SMART launch opens the app from an EHR context; CDS Hooks can return readiness and stalled-evaluation cards.
- The same standards-based integration pattern can be configured for participating EHR environments during a pilot.

DEVELOPMENT ROADMAP

M1–M2 **Prepare**

Deploy the pilot environment, finish mobile polish, and configure center roles and workflows.

M3–M4 **Onboard**

Train Center 1 staff, connect test data, and run donor and coordinator usability sessions.

M5–M6 **Pilot**

Launch a limited pilot; monitor engagement, stalled evaluations, NLDAC steps, and follow-up completion.

M7–M12 **Expand**

Add Center 2, localize key journeys, publish results, and release the deployment playbook.

PILOT PLAYBOOK

1 **Configure**

Center, user roles, workflows, and outcome baseline.

2 **Train**

Coordinator walkthrough, quick guide, and donor invitation flow.

3 **Measure**

Track four pilot targets monthly and refine the workflow.

DEPLOYMENT MODEL Cloud-hosted web application • managed PostgreSQL • no specialized hardware • one center configuration at a time

12-MONTH GOAL: demonstrate repeatable deployment at two centers and measure donor and workflow outcomes.